MINUTES OF MEETING (MOM)
Team 8
3D Printing Cost Estimation Platform

Meeting Details

Project:	3D Printing Cost Estimation Platform
Week:	Week 6
Date:	Thursday, February 27, 2026
Time:	4:00 PM IST
Location:	CIE, Ground Floor
Client:	Anil Kumar Upadhyay, Five Fingers Innovative Solutions
Client Email:	`aku1974india@gmail.com`
Client Phone:	9949497560

Team Members Present

- Abhiroop Pullepu
- Mehrish Khan
- Ronith Menneni
- Sai Hasini
- Ruschita Guddeti

Notes

- This meeting was held one day before the planned Release 1 demo (v1.0.0) to review readiness and close remaining bugs.
- Team walked through the full upload → slice → view → estimate flow using Docker Compose.
- Focus was on stabilising the application and finalising the demo script/video.

1 Release 1 Readiness Review

1.1 End-to-End Workflow Status

- **File Upload & Validation**
  - POST `/api/upload` working for STL and STEP with 50 MB limit.
  - Validation errors correctly returned for unsupported formats and oversized files.
- **3D Viewer**
  - Three.js viewer renders STL models with rotate/zoom/pan.

- Known limitation: STEP preview still pending conversion pipeline (documented as a known issue).

- **Slicing & G-code**
  - Basic slicing endpoint (`/api/slice/{job_id}`) operational using the current Trimesh-based slicer.
  - G-code-like toolpath preview available; full CuraEngine integration deferred beyond Release 1.

- **Cost Estimation**
  - Cost calculation engine computes material cost, machine time cost, and applies waste/failure factors.
  - Cost breakdown is shown on the frontend in INR with proper rounding.

1.2 Demo Script and Video

- Agreed demo flow: login → upload STL → 3D view → slice with parameters → G-code preview → cost estimate.
- Pre-recorded demo video (approx. 3:30) prepared and tested at 1.5x speed for the presentation.
- Slides updated to match the implemented features and Release 1 scope.

2 Bug Review and Fixes

2.1 Backend Bugs

- **Issue: Cost rounding inconsistencies**
  - Symptom: Total cost on the UI sometimes differed by 1 INR from the sum of components due to floating point rounding.
  - Fix: Standardised monetary rounding in the backend to two decimal places before sending the response.
  - Status: Fixed and retested on sample jobs.

- **Issue: G-code preview offset**
  - Symptom: Initial layer preview appeared slightly off-centre on the grid.
  - Fix: Applied bounding-box centring for parsed toolpath coordinates before building line segments.
  - Status: Fixed; remaining tiny visual offset accepted as non-blocking for Release 1.

2.2 Frontend Bugs

- **Issue: Multiple upload warning not cleared**
  - Symptom: After a failed upload (wrong file type), the error banner persisted even when the next upload succeeded.
  - Fix: Reset error state on successful upload; added regression test.
  - Status: Fixed.
- **Issue: Spinner not stopping on slicing failure**
  - Symptom: Slicing progress spinner continued indefinitely if backend returned a failure.
  - Fix: Unified success/failure handling in the slicing thunk; always clears loading state.
  - Status: Fixed.

3 Release Checklist Confirmation

- Docker Compose brings up all services (frontend, backend, PostgreSQL, slicing service) without errors.
- Backend and frontend unit tests passing on CI.
- Basic integration tests (upload, slice, estimate) run successfully on the release branch.
- README and setup guide updated for local deployment.
- Known issues documented (STEP preview, large model performance).

4 Next Steps

4.1 Team Deliverables (Due: February 28, 2026)

- Conduct full demo rehearsal in the morning with projector at CIE.
- Freeze code on release branch except for P0 bug fixes.
- Tag Release 1 (`v1.0.0`) after final verification and update release notes.

4.2 Client Expectations

- Live demo of the end-to-end workflow during the Release 1 presentation.
- Short discussion on priorities for the next phase (security hardening, printability analysis, email notifications).

Minutes prepared by: Team 8
Date: February 27, 2026